1. The value of x for which the angle between $\vec{a} = 2x^2\hat{i} + 4x\hat{j} + \hat{k}$ and $\vec{b} = 7\hat{i} - 2\hat{j} + x\hat{k}$ is obtuse and the angle between $\vec{b}$ and the z-axis is acute and less than $\pi/6$ is

1) $a < x < 1/2$ **2)** $1/2 < x < 15$ **3)** $x > 1/2$ or $x < 0$ **4)** ϕ

2. Three vectors $\hat{i}+\hat{j}$, $\hat{j}+\hat{k}$ and $\hat{k}+\hat{i}$ taken two at a time form three planes. The three unit vectors drawn perpendicular to these three planes form a parallelepiped of volume

1) 1/3 **2)** 4 **3)** $(3\sqrt{3})/4$ **4)** $4/3\sqrt{3}$

3. If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors such that $\vec{a}.\,\vec{b} = 0 = \vec{a}.\,\vec{c}$ and the angle between $\vec{b}$ and $\vec{c}$ is $\pi/3,$ then the value of $\left|\vec{a} \times \vec{b} - \vec{a} \times \vec{c}\right|$ is

1) 1/2 **2)** 1 **3)** 2 **4)** 1/3

4. The volume of the cuboid having combined equations of opposite faces $x^2-3x-4=0, y^2-5y+6=0$ and $z^2+6z+8=0$ is

1) 20 cu. units **2)** 10 cu. units **3)** 15 cu. units **4)** 24 cu. units

5. Let A(1, 1, 1), B(2, 3, 5) and C(-1, 0, 2) be three points, then equation of a plane parallel to the plane ABC which is at distance 2 is

1) $2x-3y+z+2\sqrt{14}=0$ 2) $2x-3y+z-\sqrt{14}=0$ 3) $2x-3y+z+2=0$ 4) $2x-3y+z-2=0$

6. The length of projection of the line segment joining the points (1, 0, -1) and (-1, 2, 2) on the plane $x + 3y - 5z = 6$ is equal to

1) 2 **2)** $\sqrt{\frac{271}{53}}$ **3)** $\sqrt{\frac{472}{31}}$ **4)** $\sqrt{\frac{474}{35}}$

7. Unit vectors $\vec{a}$ and $\vec{b}$ are perpendicular, and unit vector $\vec{c}$ is inclined at an angle θ to both $\vec{a}$ and $\vec{b}$. If $\vec{c} = \alpha\vec{a} + \beta\vec{b} + \gamma(\vec{a} \times \vec{b})$, then

1) $a = \beta$ **2)** $\gamma^2 = 1 - 2\alpha^2$ **3)** $\gamma^2 = -cos2\theta$ **4)** $\beta^2 = \frac{1+cos2\theta}{2}$

8. If vectors $\vec{b} = (tan\alpha, -1, 2\sqrt{sin\alpha/2})$ and $\vec{c} = \left(tan\ \alpha, tan\ \alpha, -\frac{3}{\sqrt{sin\alpha/2}}\right)$ are orthogonal and vector $\vec{a} = (1, 3, sin2\alpha)$ makes an obtuse angle with the z-axis, then the value of α is

1) $\alpha = (4n+1)\pi + tan^{-1}2$ **2)** $\alpha = (4n+1)\pi - tan^{-1}2$ **3)** $\alpha = (4n+2)\pi + tan^{-1}2$

4) $\alpha = (4n+2)\pi - tan^{-1}2$

9. A parallelogram is constructed on vectors $\vec{a} = 3\vec{\alpha} - \vec{\beta}, \vec{b} = \vec{\alpha} + 3\vec{\beta}$ if $|\vec{\alpha}| = |\vec{\beta}| = 2$, and angle between $\vec{\alpha}$ and $\vec{\beta}$ is $\frac{\pi}{3}$, then the length of a diagonal of parallelogram is

1) $4\sqrt{5}$ **2)** $4\sqrt{3}$ **3)** $4\sqrt{7}$ **4)** $4\sqrt{2}$

10. The equation of the plane which is equally inclined to the lines $\frac{x-1}{2}=\frac{y}{-2}=\frac{z+2}{-1}$ and $\frac{x+3}{8}=\frac{y-4}{1}=\frac{z}{-4}$ and passing through the origin is/are

1) $14x-5y-7z=0$ **2)** $2x+7y-z=0$ **3)** $3x-4y-z=0$ **4)** $x+2y-5z=0$

11. A rod of length 2 units whose one end is (1, 0, -1) and other end touches the plane $x-2y+2z+4=0,$ then

1) the rod sweeps the figure whose volume is π cubic units

2) the area of the region which the rod traces on the plane is 2π

3) the length of projection of the rod on the plane is $\sqrt{3}$ units

4) the centre of the region which the rod traces on the plane is $\left(\frac{2}{3}, \frac{2}{3} \frac{-5}{3}\right)$

12. Let $P_1 : 2x + y - z = 3$ and $P_2 : x + 2y + z = 2$ be two planes. Then, which of the following statement(s) is (are) TRUE ?

1) The line of intersection of P_1 and P_2 has direction ratios 1, 2, -1

2) The line $\frac{3x-4}{9} = \frac{1-3y}{9} = \frac{z}{3}$ is perpendicular to the line of intersection of P_1 and P_2

3) The acute angle between P_1 and P_2 is 60^0

4) If P_3 is the plane passing through the point (4, 2, -2) and perpendicular to the line of intersection of P_1 and P_2, then the distance of the point (2, 1, 1) from the plane P_3 is $\frac{2}{\sqrt{3}}$

13.

If $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}, \vec{b}=b_1\hat{i}+b_2\hat{j}+b_3\hat{k}, \vec{c}=c_1\hat{i}+c_2\hat{j}+c_3\hat{k}$ and $[3\vec{a}+\vec{b}\ 3\vec{b}+\vec{c}\ 3\vec{c}+\vec{a}]=\lambda\begin{vmatrix}\vec{a}.\hat{i} & \vec{a}.\hat{j} & \vec{a}.\hat{k}\\ \vec{b}.\hat{i} & \vec{b}.\hat{j} & \vec{b}.\hat{k}\\ \vec{c}.\hat{i} & \vec{c}.\hat{j} & \vec{c}.\hat{k}\end{vmatrix}$, then find the value of $\frac{\lambda}{4}$.

14. Let $\overrightarrow{OA} = \vec{a}, \overrightarrow{OB} = 10\vec{a} + 2\vec{b}$ and $\overrightarrow{OC} = \vec{b}$, where O,A and C are non-collinear points. Let p denote the area of quadrilateral OACB and let q denote the area of parallelogram with OA and OC as adjacent sides. If p=kq, then find k

15. If $\vec{a}$ and $\vec{b}$ are vectors in space given by $\vec{a} = \frac{\hat{i}-2\hat{j}}{\sqrt{5}}$ and $\vec{b} = \frac{2\hat{i}+\hat{j}+3\hat{k}}{\sqrt{14}}$, then find the value of $(2\vec{a}+\vec{b}).\,[(\vec{a}\times\vec{b})\times(\vec{a}-2\vec{b})]$

16. Let the equation of the plane containing line $x-y-z-4=0=x+y+2z-4$ and parallel to the line of intersection of the planes $2x+3y+z=1$ and $x+3y+2z=2$ be $x+Ay+Bz+C=0$. Then find the value of $|A+B+C-4|$.

17. The distance of the point P(-2, 3, -4) from the line $\frac{x+2}{3}=\frac{2y+3}{4}=\frac{3z+4}{5}$ measured parallel to the plane $4x+12y-3z+1=0$ is d, then find the value of (2d - 8).

18. A variable plane at a distance of 1 unit from the origin cuts the coordinate axes at A, B and C. If the centroid $D(x, y, z)$ of triangle ABC satisfies the relation $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = k,$ then the value of k is ______.

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